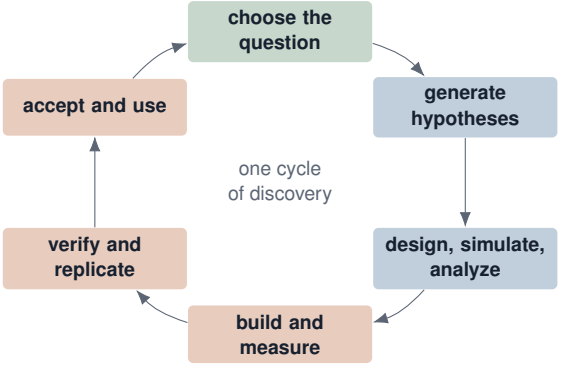
Research 2.0

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Research is a loop, and only some of its arrows got cheaper



Blue: made of symbols. Text, code, search. AI makes these 10–1000× cheaper.

Copper: made of contact with the world. Samples, instruments, time, other people's trust. AI barely touches them.

Green: made of judgment. Which question is worth a cycle at all.

Research 2.0 is what happens to a loop when half its arrows become nearly free. The speed of the loop is set by the arrows that did not.

The thesis in three lines

Boosted

Everything made of symbols and checked by a fast verifier: reading, hypothesizing, searching, coding, simulating, proving, writing.

Bottleneck

Everything made of contact with the world: experiments, prototypes, field data, replication — and the human verification of an exploding volume of claims.

Moat

Moves from *knowing* to *measuring*: from expertise and technique to ground-truth facilities, unique data, trusted verification, and taste.

> **When ideas become abundant, ground truth becomes the currency.** Value moves to whoever can say, credibly, which ideas are true.

Stage by stage: what AI accelerates

Stage	What AI does	Verifier	Boost
Reading and synthesis	Reads thousands of papers; maps a field in hours	Citations, expert	■■■■■
Coding and analysis	Writes, tests, and runs analysis pipelines	Tests, reruns	■■■■■
Writing and grants	Drafts papers, proposals, reports	The reader	■■■■■
Hypothesis generation	Proposes many candidates cheaply	An experiment	■■■■□
Design and search	Searches huge design spaces with simulation in the loop	Simulation	■■■■□
Simulation	Learned surrogates; faster solvers	Measured data	■■■□□
Build and measure	Self-driving labs, in narrow domains	Nature	■■□□□
Field validation, reliability	—	Time	■□□□□
Acceptance by peers	—	People	■□□□□

Boost falls as you go down the table: from symbols, to models, to matter, to time, to people.

The six bottlenecks of Research 2.0

Bottleneck	Why AI cannot remove it	Partial relief
1 **Physical experiment**	Nature answers at its own rate; samples must be made	Automation, parallel rigs
2 **Verification and trust**	Checking needs independent data and accountable people	Formal proofs, open data
3 **Ground-truth data**	Unique, physical, longitudinal; cannot be scraped	Instrumented fleets
4 **The question**	Value depends on goals outside the data	Faster user feedback
5 **Facilities, compute**	Instruments, fabs, and frontier compute stay expensive	Shared national facilities
6 **Human absorption**	Someone must understand, trust, and act on results	Better artifacts

All six are made of matter, time, or people. None is made of symbols — which is why AI moved them to the front of the queue.

The moat, 1.0 → 2.0

Eroding

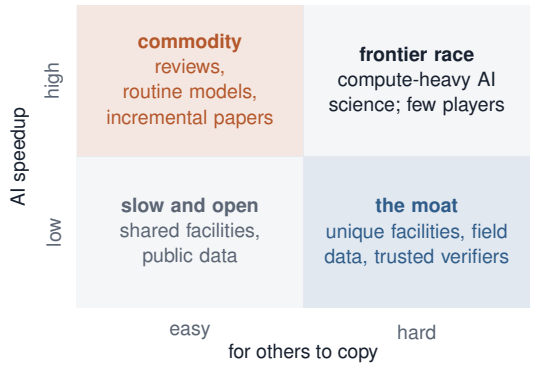
- **Mastery of the literature** — now a query.
- **Methods and code** — simulation, analysis, and modeling skill become shared tools.
- **Writing and grant craft** — polish is free.
- **Workforce size** — labor for symbolic tasks is no longer scarce.
- **Access to software and papers** — largely gone as a barrier.

Strengthening

- **Ground-truth facilities** — instruments, fabs, test beds, field sites.
- **Proprietary data** — especially failure and field data.
- **Trusted verification** — benchmarks, standards, a record of being right.
- **Taste** — choosing the question worth a cycle.
- **Real problems** — partners who deploy, and say what matters.
- **Loop speed** — verified cycles per year.

> **The moat moves from what is in your head to what is in your lab, your data, and your track record.**

The paradox: AI boosts most what it commoditizes most



The tasks AI speeds up most are the tasks *anyone* with AI can now do. A 20× speedup that everyone gets is a 1× advantage.

The tasks AI speeds up least — measurement, field data, verification — are exactly the ones others cannot copy.

Advantage = your speedup relative to the field, not relative to your past self.

Compete where AI is slow; use AI everywhere else. The moat is in the lower-right quadrant.

Institutions: the K-shape of Research 2.0

Rising

- **Owners of frontier compute** — in AI research, industry built 96% of the largest models; ∼70% of AI PhDs now go to industry (Ahmed *et al.*, 2023).
- **Owners of physical verifiers** — national labs, test centers, instrumented fleets.
- **Owners of unique data** — hospitals, utilities, manufacturers, operators.

Squeezed

- **Desk-research groups** doing incremental symbolic work.
- **Journals as gatekeepers** — volume overwhelms review.
- **Credentialing knowledge** — what a degree certified is now a query.

The university's durable moat

Neutral verification, long horizons, shared facilities, and training people to judge.

> **Universities lose their monopoly on knowing. They can keep the role of trusted verifier** — if they invest in it.

The researcher: from producer to director and verifier

Depreciating skills	Appreciating skills
Recalling the literature	Choosing questions worth a verification cycle
Writing code and analysis	Designing the decisive experiment
Deriving routine models	Knowing where the model breaks
Writing polished prose	Skepticism: finding the 1-in-5 wrong claim
Managing a large team of junior labor	Directing many AI agents, and checking them
Publishing many papers	Building a record of being right

> **The scarce researcher of 2.0 is the one who can tell a true result from a plausible one** — fast, and in their own domain.

Research 1.0 → 2.0

	Research 1.0	Research 2.0
Scarce input	Knowledge and skilled labor	Ground truth and judgment
Binding cost	Absorbing the literature	Verifying the candidates
Unit of output	The paper	The verified artifact
Quality control	Peer review, after the fact	Verifiers attached to every claim
Lab structure	PI, students, postdocs	PI, students, agents, instruments
Measure of a lab	Papers and citations	Verified results, reused
Moat	Expertise and technique	Facilities, data, trust, taste

The new unit of output: a claim with its verifier attached

Claim	1.0 output	2.0 output
Theorem	Proof in a paper	Machine-checked proof (Lean)
Algorithm	Description and a table	Code, tests, benchmark — rerunnable
Simulation	Figures	Pipeline plus validation against measured data
Experiment	Paper	Raw data, protocol, uncertainty — replicable
Hardware	Paper and photos	Design files plus test report to a standard

> **In 1.0 a paper was the result. In 2.0 it is a pointer to the result.** Readers — human or AI — should be able to check, not just believe.

Power electronics research 2.0

Boosted	Bottleneck	Moat
Topology and modulation search	Prototype build and test hours	High- and medium-voltage test facilities
Magnetics and thermal co-design	Insulation, partial discharge, arcing	Long-term reliability and field-failure data
Control tuning and firmware	Lifetime and reliability — years of aging	Device-to-system test beds
PCB layout, EMI pre-compliance	Field data from real fleets	Partnerships with operators who deploy
Literature, simulation, surrogates	Certification to standards	A seat where the standards are written

Design will be cheap. Knowing how long it lasts in the field, will not. Reliability is the field's slowest verifier — and its deepest moat.

CONCLUSION

Abundant ideas, scarce ground truth

Boosted

Reading, hypothesizing, designing, coding, simulating, proving, writing — by 10–1000×, where the verifier is fast.

Bottleneck

Experiment, field data, replication, review, trust.

Moat

From knowing to measuring: facilities, unique data, trusted verification, taste.

> **Research 1.0 was limited by what we could know. Research 2.0 is limited by what we can verify.**

References

- T. Tao, *Math 2.0*, Caltech, Oct. 2026.

Thank you

Comments, critiques, and counterexamples are welcomed.

Companion to *Physics 2.0*, *Engineering 2.0*, *Society 2.0*, and *Entrepreneurship 2.0*.

AI was used in preparing these slides.